

Fertility Project Experiment Note: Model Setup Trial + Empirical Idea Ranking

Project 03: Fertility and Housing Supply

February 20, 2026

Purpose

This note does two things:

1. Runs a small computational experiment based on the model setup in `projects/03_Fertility_and_Housing_Supply/Notes/2026-02-19_fertility_extension_model_key_equations.tex`.
2. Ranks empirical causal ideas and expands the top 2 in detail (including a NIMBY-focused design).

1 Part I: Model setup trial (code + run)

What was implemented

A version-0 transition simulation was coded in:

- `projects/03_Fertility_and_Housing_Supply/Code/fertility_extension_experiment.py`

The prototype keeps the model intentionally stylized but consistent with the note's key feedback loop:

prices $\rightarrow$ fertility $\rightarrow$ cohort shares $\rightarrow$ political tightness $\rightarrow$ prices.

Scenarios simulated for 50 periods:

- No damping (aggressive update).
- Damping 0.25 (stabilized fixed-point style update).
- Damping 0.25 + family-sized supply policy shift from period 15 onward.

Outputs were written to:

- `projects/03_Fertility_and_Housing_Supply/Notes/build/model_experiment_summary.csv`
- `projects/03_Fertility_and_Housing_Supply/Notes/build/model_experiment_paths.png`

Run command

```
python3 projects/03_Fertility_and_Housing_Supply/Code/fertility_extension_experiment.py
```

Results (from summary csv)

Scenario	Avg price (t=10..49)	Avg fertility (t=10..49)	Max abs price change
No damping	-5.939	0.723	2.426
Damping 0.25	-2.597	0.416	1.948
Damping 0.25 + policy	-2.638	0.443	1.349

Interpretation:

- The undamped update is much less stable (larger jumps and corner-like dynamics).
- Damping materially improves numerical behavior.
- The family-sized supply shift raises fertility in the damped run (0.416 to 0.443 average fertility rate in this toy environment).

What failed and what was changed: Initial runs generated overflow in exponential terms (numerical blow-up), so the code was patched with bounded exponent arguments and price clamps. This is direct evidence that the full model will require strong numerical safeguards (damping, normalization, bounded controls).

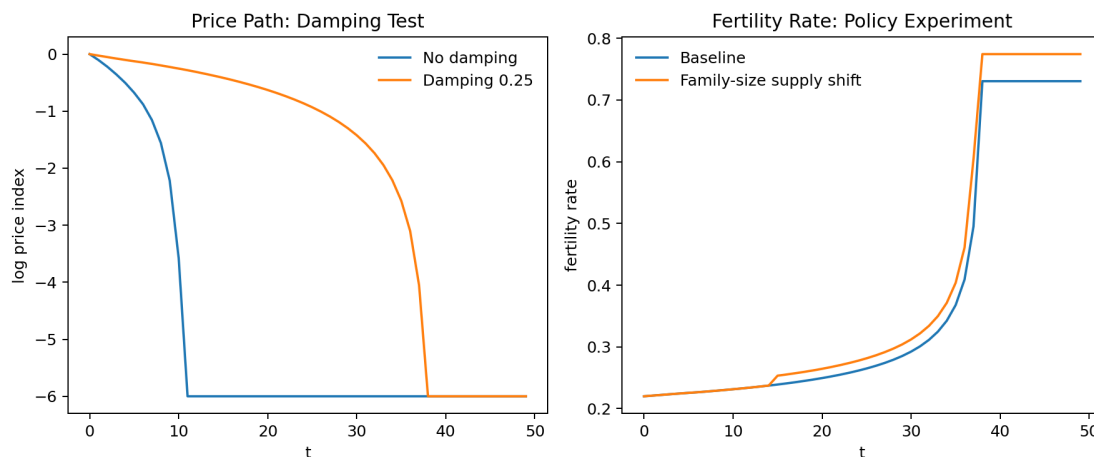


Figure 1: Version-0 paths: damping sensitivity and policy experiment

Value of this experiment

This is not structural estimation. It is a feasibility stress test that confirms the convergence warning in the model note is operationally relevant, not just theoretical.

2 Part II: Empirical causal ideas ranked

Ranking criteria

Scores reflect four dimensions: identification credibility, data feasibility, novelty potential, and alignment with your housing-fertility channel.

Idea	Rank
State middle-housing legalization (staggered reform) and fertility	1
NIMBY political-economy design: close local elections and pro-housing wins	2
Family-sized unit-composition shock and fertility (shift-share IV)	3
Short-term rental crackdown and fertility	4
Single-city zoning mega-reform with legal interruptions	5

3 Top idea 1 (expanded): State middle-housing legalization and fertility

Question: Do legalizations of duplex/triplex/fourplex construction increase births among prime childbearing cohorts?

Estimand: Dynamic ATT of reform exposure on age-specific birth rates.

Design: Staggered DiD/event study at county or PUMA level with intensity-by-exposure.

Core specification (event study):

$$BirthRate_{it} = \sum_{k \neq -1} \beta_k \cdot 1\{EventTime_{it} = k\} + \alpha_i + \delta_t + X'_{it}\Gamma + \varepsilon_{it}.$$

Intensity extension:

$$BirthRate_{it} = \beta (Post_{it} \times Exposure_i) + \alpha_i + \delta_t + X'_{it}\Gamma + \varepsilon_{it}.$$

Data (feasible now):

- CDC WONDER natality (county-level births by mother's age).
- Census Building Permits Survey (new units; single vs multifamily).
- FHFA/Zillow house-price and rent controls.
- Reform timing panel (state/city policy adoption and implementation).

Why this is strong:

- Direct policy shock on the supply margin.
- Directly matches your mechanism (housing constraints $\rightarrow$ family formation/fertility).
- Good external validity due to multi-jurisdiction rollout.

Main threats and tests:

- Threat: pre-trends in treated areas.
- Test: pre-period event coefficients and placebo reform dates.
- Threat: migration composition shifts.
- Test: control for migration proxies; run residence-duration restricted samples if possible.
- Threat: weak first stage on effective supply.
- Test: document permit/unit-mix response as first-stage evidence.

Novelty claim (current confidence): Medium-High. The novelty is to estimate fertility effects of explicit middle-housing legalizations, not only generic price shocks.

4 Top idea 2 (expanded): NIMBY political-economy design via close elections

Question: Does narrowly electing pro-housing local governments causally increase supply and, through this channel, affect fertility?

Estimand:

- First stage: effect of pro-housing win on permits/unit composition and prices.
- Second stage: effect of induced housing-supply changes on fertility.

Design: Close-election RD at municipal level + 2SLS mediation.

First stage:

$$Supply_{it} = \pi ProHousingWin_{it} + f(Margin_{it}) + \eta_i + \tau_t + u_{it}.$$

Second stage:

$$BirthRate_{it} = \beta \widehat{Supply}_{it} + f(Margin_{it}) + \eta_i + \tau_t + e_{it}.$$

Why this is NIMBY-relevant and novel:

- It directly uses political turnover in the local anti-/pro-development margin.
- It maps your theoretical political channel into a causal empirical design.
- Compared with standard housing papers, the fertility endpoint is a newer margin.

Data requirements (hard but feasible):

- Municipal election returns with candidate coding (pro-housing stance).
- Municipal permits and unit mix.
- Local prices/rents.
- Birth outcomes at the finest feasible geography (county minimum; subcounty where available).

Main risks:

- Candidate stance coding may be noisy.
- Fertility outcomes may be coarser than election geography.
- Sample size can be thin once restricting to close races.

Risk mitigation:

- Use transparent text-based coding protocol with inter-rater checks.
- Aggregate to county-year with municipality exposure weights.
- Pool multiple states and cycles.

Novelty claim (current confidence): Medium. Potentially high upside, but novelty depends on whether recent housing-politics papers already run near-identical RD-to-fertility pipelines.

5 Pragmatic recommendation

If the objective is “credible result in finite time,” start with Top 1. If the objective is “high-upside NIMBY contribution,” develop Top 2 in parallel as a second track.

Deliverables generated in this session

- Prototype code: `projects/03_Fertility_and_Housing_Supply/Code/fertility_extension_experiment.py`
- Experiment outputs: `projects/03_Fertility_and_Housing_Supply/Notes/build/model_experiment_*`
- This report PDF/TeX.